PIB Synthesizer
Analog & Digital Inputs

Introduction

This project involves the creation of a 25-key keyboard (2 octaves) with added functionality to read inputs from an external MIDI keyboard. Inputs from both keyboards are processed by a microcontroller, which translates key presses into corresponding voltage signals sent to the VCO module to control pitch. Each key press also generates a Gate signal, triggering the ADSR module to define the sound's envelope.

Objective

The analog input has the following objectives assigned to it: to develop a functional keyboard with a minimum range of one octave, capable of producing the sound of up to four octaves. The digital input also has objectives: to decode a MIDI signal from a MIDI keyboard, as provided by the client.

Requirements

Functional requirements for both inputs:

- When a new key is pressed, the previous tone stops and the new tone begins.
- A gate signal must be sent whenever a key on the keyboard is pressed or released. This signal is 0V when off and 3.3V when on.

Functional requirements for Analog input:

- A minimum of 1 octave of keys, with the ability to play 4 octaves.
- An analog signal for pitch.
- 1 octave should have a range of 1 volt.

Functional requirements for digital input:

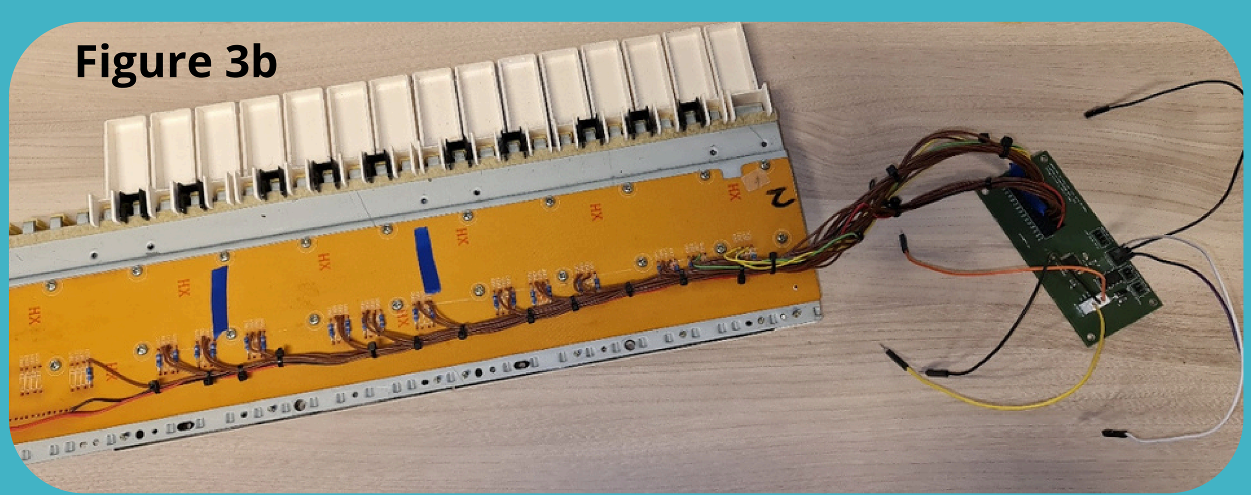
- <u>The incoming byte from the MIDI keyboard is read, and the corresponding pitch is sent.</u>
- The MIDI input must comply with the NEN-EN-INC 63035 protocol for the most part.

Non-functional requirements for both inputs:

- No velocity sensitivity.
- A maximum cost of 50 euro.
- The module must be compatible with other modules within the synthesizer.
- It must be safe and easy to use.

Figure 3a

Figure 3b

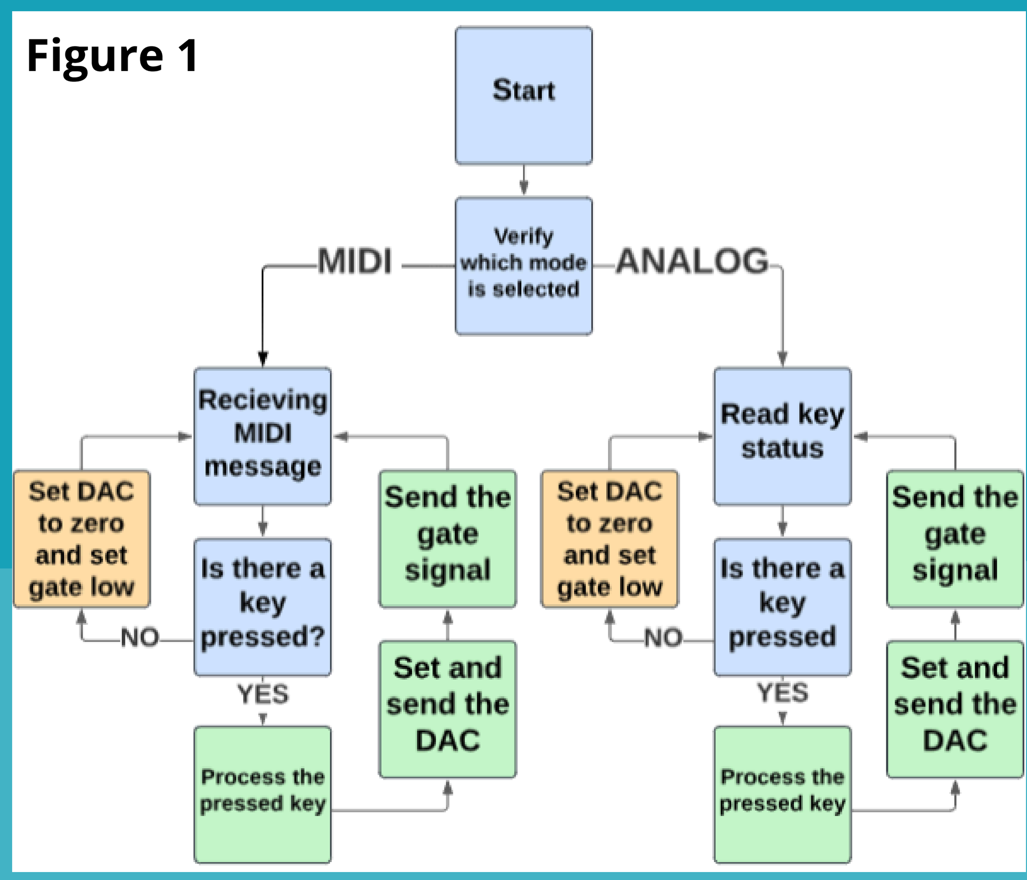


Software

The built-in keyboard checks the status of each key using polling. When a key is pressed, a 10-bit DAC (Digital-to-Analog Converter) is triggered to produce a specific voltage on an output pin. At the same time, a gate signal is sent to indicate that a key has been pressed.

Meanwhile, for the MIDI keyboard, two bytes are read: the first byte contains pitch information, and the second byte represents the gate signal. Based on this data, the 10-bit DAC generates the corresponding voltage, and a gate signal is sent.

Figure 1

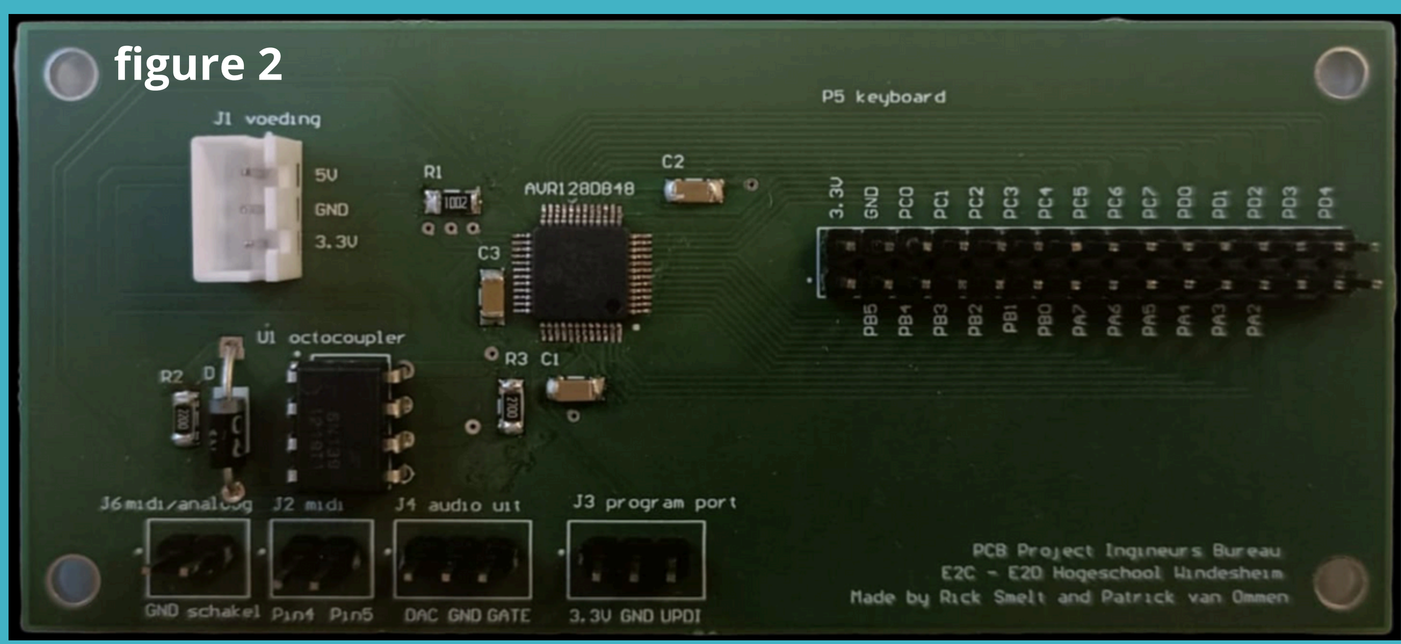


Hardware

The custom PCB (figure 2) was designed in Altium, featuring the AVR128DB48 microcontroller at its core, which controls all functionalities within the program. A standard circuit has been integrated for receiving MIDI messages, ensuring essential connectivity. Additionally, a row of pin headers has been included for interfacing with the analog keyboard. To optimize power stability, strategically placed decoupling capacitors have been used.

The analog keyboard (shown in Figures 3a and 3b) was a pre-existing keyboard. A PCB already connected to the keys was utilized, along with the keys themselves. These components were integrated into the new housing and connected accordingly.

figure 2



Conclusion

The project consisted of several components, including an Elektor article, a poster, and a personal PCB design. Although the underlined requirements in the requirements list were not fulfilled, all other requirements were successfully met and thoroughly tested.

Finally, the following questions were addressed:

1. **What voltages should be used to send a stable signal to the oscillator, sequencer, and ADSR module?**
   - A DAC voltage between 1 and 3 volts is sent, along with a gate signal of 3.3 volts.
2. **What does the analog signal look like, and how is it processed?**
   - The analog signal ranges between 1 and 3 volts, determined by the pressed button. Simultaneously, a gate signal is sent: 3.3 volts when high and 0 volts when low.
3. **What does the digital signal look like, and how is it processed?**
   - The MIDI signal contains three incoming bytes: the first byte represents the channel, the second byte specifies the key pressed, and the third byte indicates touch sensitivity (gate).

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